

ABET Self-Study Evaluation and Continuous Improvement Report

Course: CHE 312 – Transport Phenomena II, Spring 2024

1. Self-Study Evaluation (Findings and Interpretation)

Student Outcome 1 (SO1) for CHE 312 was assessed using the comprehensive final exam as a direct measure of students' ability to apply principles of heat and mass transfer to solve complex engineering problems.

The **overall class performance was significantly below expectations**, with an **average exam score of 41.67** and a **standard deviation of 27.17**. These results indicate both **low overall achievement** and **substantial variability in student performance**, suggesting that many students did not develop the necessary competencies in key course topics such as conduction, convection, diffusion, and transport correlations.

The **performance indicator average of 1.88 (on a 3-point rubric scale)** falls **below the attainment benchmark of 2.0**, indicating that **SO1 is not attained** for this course offering. The distribution of student performance further reinforces this conclusion:

- **Satisfactory (S): 7 students**
- **Developing (D): 7 students**
- **Unsatisfactory (U): 10 students**

A majority of students (**17 out of 24, ~71%**) did not reach satisfactory attainment, with a large proportion performing at the unsatisfactory level.

This outcome highlights significant deficiencies in students' ability to:

- Formulate and solve transport equations for heat and mass transfer systems
- Apply boundary conditions and simplifying assumptions appropriately
- Integrate multiple concepts (e.g., coupled heat and mass transfer phenomena)
- Interpret physical meaning and engineering implications of solutions

The **high standard deviation** further suggests a wide disparity in student preparedness and learning, potentially due to differences in prerequisite knowledge, engagement, or effectiveness of instructional delivery.

Based on instructor reflection and assessment results, several contributing factors have been identified:

- **Misalignment between instruction and assessment:** Students may not have had sufficient exposure to the level of complexity presented in the final exam.
- **Insufficient mastery of prerequisite knowledge:** Weaknesses in prior coursework (e.g., CHE 311, differential equations) likely impacted student performance.
- **Limited formative assessment:** Lack of frequent, low-stakes assessments reduced opportunities for early feedback and course correction.
- **Cognitive overload:** The abstract and mathematically intensive nature of heat and mass transfer may have overwhelmed students without sufficient scaffolding.
- **Passive learning environment:** Traditional lecture-based delivery may not have sufficiently engaged students in active problem-solving practice.

2. Continuous Improvement Actions

To address the identified deficiencies and improve attainment of SO1, the following actions will be implemented:

A. Course Content and Alignment

- Conduct a **systematic review of alignment** between learning objectives, instructional materials, and assessment instruments.
- Ensure that **exam problems reflect practiced problem types and difficulty levels** encountered in lectures and assignments.

B. Increased Formative Assessment

- Introduce **weekly quizzes and structured homework assignments** targeting key concepts such as conduction models, convective coefficients, and mass diffusion.
- Use **formative assessments early and throughout the semester** to identify gaps in understanding before high-stakes evaluations.

C. Scaffolded Problem-Solving Instruction

- Break down complex problems into **step-by-step solution frameworks**, emphasizing:
 - Physical interpretation
 - Assumptions and boundary conditions
 - Mathematical formulation
- Incorporate **guided examples and partially completed problems** to support learning progression.

D. Active Learning Strategies

- Implement **in-class problem-solving sessions**, peer instruction, and think-pair-share activities to increase student engagement.
- Use **conceptual questions and real-world applications** to strengthen understanding of abstract transport phenomena.

E. Targeted Student Support

- Offer **supplemental instruction sessions, tutoring, and review workshops**, particularly before exams.
- Develop **early warning mechanisms** (e.g., performance tracking on quizzes) to identify and support at-risk students.

F. Assessment and Rubric Refinement

- Reevaluate **exam design** to ensure clarity, appropriate difficulty, and alignment with course objectives.
 - Refine the **rubric criteria** to clearly distinguish between performance levels and ensure consistent evaluation.
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3. Reassessment Plan

The effectiveness of these improvement measures will be evaluated in the next offering of CHE 312 by:

- Monitoring **increases in average exam scores and rubric performance indicators**
- Reducing the **percentage of unsatisfactory ratings**
- Decreasing the **standard deviation** to reflect more consistent student achievement

- Analyzing **student performance on specific topics** (e.g., diffusion, convection correlations)

4. Conclusion

The current assessment indicates that **Student Outcome 1 is not attained** in CHE 312. The results reveal significant challenges in student understanding and application of heat and mass transfer principles. The identified continuous improvement actions focus on enhancing alignment, increasing formative feedback, strengthening problem-solving instruction, and supporting at-risk students. These changes are expected to substantially improve student performance and outcome attainment in future offerings of the course.